SPREADING GEOMETRIC IRREDUCIBILITY FROM A GEOMETRICALLY IRREDUCIBLE GENERIC FIBRE TO A NONEMPTY OPEN OF THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 7556-7567 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 7569-7641, the complete author proof (71 body lines).

Editorial note. Navigation stubs: 10 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Geometric irreducibility and geometric reducedness of fibres, separable finitely generated field extensions with a transcendence basis, and the birational common-open lemma for schemes with isomorphic fraction fields; these are cited inputs, while the model construction in the retained span is the author's own.

Source reference. Original lemma 030W: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7641–7641. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 030Q: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7641–7641. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0555: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7432–7440. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0557: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7492–7500. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0550: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6719–6739. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 054X: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6524–6530. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 054Z: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6690–6694. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0552: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7641–7641. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 054Q: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7641–7641. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 035W: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7641–7641. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. Assume*

(1) *Y is irreducible with generic point η,*
(2) *X_η is geometrically irreducible, and*
(3) *f is of finite type.*

Then there exists a nonempty open subscheme $V \subset Y$ such that $X_V \to V$ has geometrically irreducible fibres.

First proof of Lemma 0.1. We give two proofs of the lemma. These are essentially equivalent; the second is more self contained but a bit longer. Choose a diagram

$$\begin{array}{ccccc} X' & \xrightarrow{g'} & X_V & \longrightarrow & X \\ {\scriptstyle f'}\downarrow & & \downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & V & \longrightarrow & Y \end{array}$$

as in Lemma 0550. Note that the generic fibre of f' is the reduction of the generic fibre of f (see Lemma 054Z) and hence is geometrically irreducible. Suppose that the lemma holds for the morphism f'. Then after shrinking V all the fibres of f' are geometrically irreducible. As $X' = (Y' \times_V X_V)_{red}$ this implies that all the fibres of $Y' \times_V X_V$ are geometrically irreducible. Hence by Lemma 0555 all the fibres of $X_V \to V$ are geometrically irreducible and we win. In this way we see that we may assume that the generic fibre is geometrically reduced as well as geometrically irreducible and we may assume $Y = \mathrm{Spec}(A)$ with A a domain.

Let $x \in X_\eta$ be the generic point. As X_η is geometrically irreducible and reduced we see that $L = \kappa(x)$ is a finitely generated extension of $K = \kappa(\eta)$ which is geometrically reduced and geometrically irreducible, see Varieties, Lemmas 035W and 054Q. In particular the field extension L/K is separable, see Algebra, Lemma 030W. Hence we can find $x_1, \ldots, x_{r+1} \in L$ which generate L over K and such that $x_1, \ldots, x_r$ is a transcendence basis for L over K, see Algebra, Lemma 030Q. Let $P \in K(x_1, \ldots, x_r)[T]$ be the minimal polynomial for x_{r+1}. Clearing denominators

we may assume that P has coefficients in $A[x_1, \ldots, x_r]$. Note that as L is geometrically reduced and geometrically irreducible over K, the polynomial P is irreducible in $\overline{K}[x_1, \ldots, x_r, T]$ where $\overline{K}$ is the algebraic closure of K. Denote

$$B' = A[x_1, \ldots, x_{r+1}]/(P(x_{r+1}))$$

and set $X' = \mathrm{Spec}(B')$. By construction the fraction field of B' is isomorphic to $L = \kappa(x)$ as K-extensions. Hence there exists an open $U \subset X$, and open $U' \subset X'$ and a Y-isomorphism $U \to U'$, see Morphisms, Lemma 0552. Here is a diagram:

$$\begin{array}{ccccccc} X & \longleftarrow & U & = & U' & \longrightarrow & X' = \mathrm{Spec}(B') \\ & \searrow & \downarrow & & \downarrow & \swarrow & \\ & & Y & = & Y & & \end{array}$$

Note that $U_\eta \subset X_\eta$ and $U'_\eta \subset X'_\eta$ are dense opens. Thus after shrinking Y by applying Lemma 054X we obtain that U_y is dense in X_y and U'_y is dense in X'_y for all $y \in Y$. Thus it suffices to prove the lemma for $X' \to Y$ which is the content of Lemma 0557. $\square$